Younghoo Henry Cho

Data & Backend Engineer | Data Platforms, Provenance, Lineage
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Summary

Data and backend engineer who builds systems that let the receiver of data check its origin and history. Currently on Connected Car Services (CCS) for Hyundai, Kia, and Genesis, where factory and carrier records must agree: Java/Spring services on Kubernetes, carrier API integrations (Verizon, Bell), and Python data platforms for data-quality monitoring. Ph.D. (2026): built DEMETER, a Hyperledger Fabric system that lets the receiver of a shared data file check its origin and recorded history without contacting earlier handlers. Since 2019: backend systems, data infrastructure, and applied machine learning; two patents (one granted, one pending) and 10 publications.

Experience

System Engineer (Backend & Data Platform), Solomon America, Duluth, GA **Sep 2026 – Present**

System Engineer, Hyundai AutoEver America

- Operate five internal data platforms (Python/Django) and 10+ automation tools for CCS data-quality monitoring, analysis, correction, and reporting, and build new tools and features on them.
- Build and operate carrier integrations (Verizon, Bell) that provision and bill vehicle connectivity: API and callback flows for line activation and deactivation, data-plan changes, and usage and billing events.
- Diagnose production incidents to root cause in the platform's Java/Spring services and ship the code fixes; find security weaknesses in the code and ship hardening changes.
- Develop and operate CCS services on Kubernetes that connect Hyundai, Kia, and Genesis vehicles in the U.S. and Canada to each brand's services: vehicle factory, carrier network, vehicle-control, and remote-command data.
- Define integration designs and scope estimates for new in-vehicle data products.

Graduate Research Assistant (Data Infrastructure), University of Florida **Aug 2022 – Aug 2026**

- Designed and built DEMETER, a data-provenance system on Hyperledger Fabric that lets an organization confirm a shared file is unchanged since registration and trace its transfers and declared source files, without contacting the organizations that handled it earlier.
- Tested DEMETER on Florida soil-carbon data against the sample-ID matching used on that data: it flagged a lab file with one changed byte, rejected a declaration citing an unregistered file, and resolved a sample ID shared by two soil horizons; sample-ID matching failed all 3 cases.
- Built the core as Go chaincode behind a Node.js/Express REST API, taking each record's author from the submitting organization's certificate, and ran the two-organization Fabric network on Docker with TLS on every node.
- Extended the ledger with transformation declarations so a combined, split, or modified file traces back to its registered sources (all 6 test scenarios passed, including rejection of 2 invalid declarations), and added a plain-English question box with keyword rules and a local LLM fallback (Phi-3 via Ollama).

AI Engineer, Billion21 **Dec 2019 – Aug 2022**

- Built computer-vision models in Python that tracked whole-tank fish feeding behavior as input to an automated feeding system and estimated individual fish size for growth monitoring.

AI Consultant, Global BioAg Linkages **Jan 2020 – Dec 2020**

- Planned AI/ML adoption for agriculture and aquaculture companies, assessing technical feasibility and market viability for each.

Education

University of Florida, Ph.D., Agricultural and Biological Engineering, GPA: 3.95/4.0 **Aug 2022 – Aug 2026**
Texas Tech University, B.S., Computer Science, Minor in Mathematics, GPA: 3.8/4.0 **Aug 2015 – Aug 2022**

Selected Publications

10 total, including under review

- Cho, Y., Yu, Z., Ampatzidis, Y., Nam, J.-S. *Facilitating a Future Agricultural Data Ecosystem: A Cross-Sectoral Review of Blockchain Application in Data Sharing*. Precision Agriculture, 27, 138 (2026).
- Cho, Y. et al. *DEMETER: An Integrated Trust Infrastructure for Shared Agricultural Data*. Under review.
- Cho, Y. et al. *An Enabling Infrastructure Layer for Origin Verification and Transfer Tracing of Shared Agricultural Data Files*. Under review.
- Cho, Y. et al. *An Infrastructure Layer for Tracing Transformed Agricultural Data Back to Its Sources*. Under review.

Patents

- *Blockchain-Based Laver Supply Data Management Method and Smart Laver Supply Chain Management System Using the Same*. KR10-2024-0011399, filed 2024, pending.
- *Methods for Selecting Spores from Laver Farm Devices*. KR102034354B1, 2019 (granted).

Skills

Languages: Python, Java, Go, SQL, JavaScript, C, C++, R
Data Engineering: Apache Airflow, Apache Spark, Apache Kafka
Databases: Oracle, PostgreSQL, MongoDB
Distributed Systems & Backend: Django (Python), Spring (Java), Node.js, Express, REST APIs, gRPC, Hyperledger Fabric
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD (GitHub Actions), Git, Linux
ML & AI: PyTorch, TensorFlow, OpenCV, YOLO, SAM, XGBoost, LightGBM, LSTM, Phi-3, LangChain, Ollama